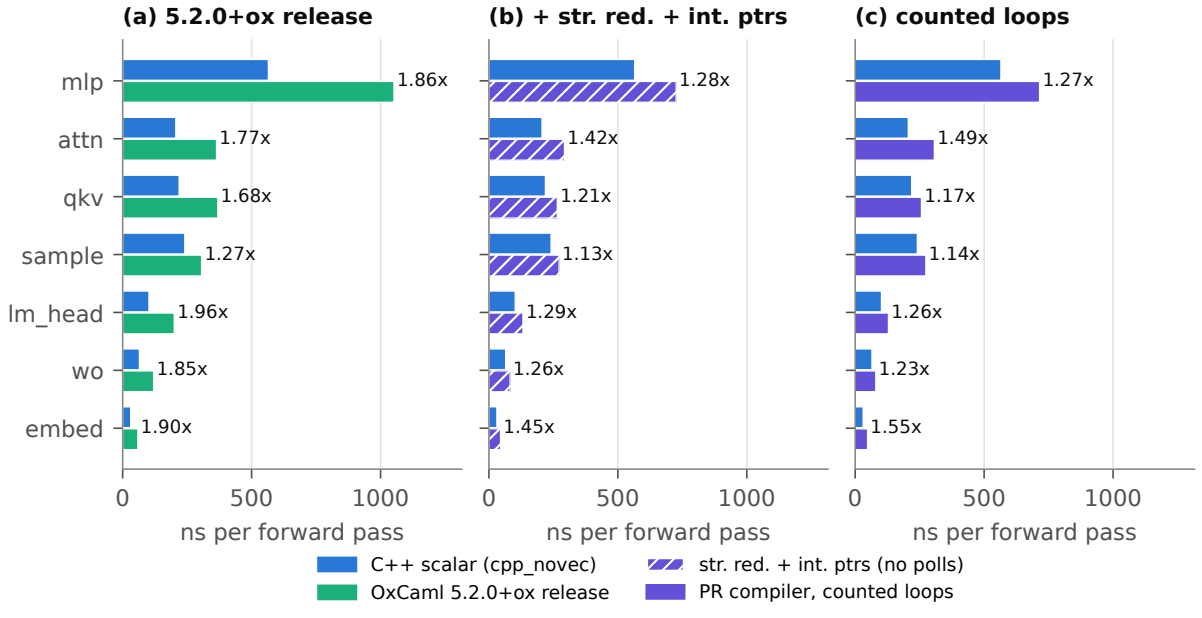
(a) 5.2.0+ox release
(b) + str. red. + int. ptrs
(c) counted loops
mlp
attn
qkv
sample
lm_head
wo
embed
1.86x
1.77x
1.68x
1.27x
1.96x
1.85x
1.90x
1.28x
1.42x
1.21x
1.13x
1.29x
1.26x
1.45x
1.27x
1.49x
1.17x
1.14x
1.26x
1.23x
1.55x
0
500
1000
ns per forward pass
C++ scalar (cpp_novec)
OxCaml 5.2.0+ox release
str. red. + int. ptrs (no polls)
PR compiler, counted loops